Daniel-John Diala

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EDUCATION

Monmouth University, West Long Branch, NJ — Expected May 2029
Bachelor of Science in Software Engineering · Honors School — **GPA:** 4.0/4.0
Honors: Dean's List (Fall 2025, Spring 2026); Academic Excellence Scholarship; Shirley Family Scholarship
Coursework: Data Structures & Algorithms, Discrete Mathematics, Computer Architecture, Calculus I & II

PROJECTS

Pulse-Net: Pandemic Early-Warning System | *Python, PyTorch, FastAPI, PostgreSQL, React* — Jan 2026 – Present

- Building an outbreak early-warning prototype where 5 simulated hospitals train PyTorch models locally and share signed, differentially private (ε = 1.0) updates, keeping patient data on site
- Designed a transformer (cross-entropy, supervised contrastive, and masked-symptom reconstruction losses) that flags possible novel pathogens; 90% held-out accuracy across 16 classes on synthetic data
- Merged updates with a norm-clipped coordinate-wise median to resist poisoning, and built encrypted aggregation with CKKS, Shamir k-of-n key escrow, zero-knowledge range proofs, and a Merkle audit log
- Built a FastAPI backend (300+ endpoints, PostgreSQL) and a React dashboard with an outbreak map, WebSocket alerts, and SEIR spread models; 4,600+ tests run in GitHub Actions CI

Sentiment Analysis Tool | *Python, spaCy, Flask, SQLite, Streamlit, Docker* — Dec 2025 – Jan 2026

- Built an NLP engine that sorts reviews as positive, negative, neutral, or mixed using a 6,800-term lexicon and negation cues
- Added aspect-based analysis and a Dockerized Flask REST API (15+ endpoints) feeding a Streamlit dashboard
- Cut runtime for 500 reviews from 17s to 5s (3.4×) via batched spaCy processing, lemma caching, and bulk SQLite inserts

EXPERIENCE

AI Literacy Teaching Assistant | Monmouth University — Sep 2026 – Present

- Help students set up agentic AI systems and use AI tools effectively and ethically
- Built a Next.js exam-prep app for the Claude Certified Architect exam (180 questions, 10 labs), deployed on AWS EC2

Undergraduate Research Assistant | Monmouth University — May 2026 – Present

- Modernized GArel, an OCaml bidirectional type checker (from Qu et al., ICFP 2019) that analyzes program cost statically
- Fixed drift in a 36-program test corpus, stress-tested the checker, and developed one of two independent solution paths
- Currently evaluating CHC- and SMT-solver-based alternatives to scale the analysis
- Built an educational web app, hosted on AWS, that visualizes constraint generation to teach bidirectional type checking

SKILLS & CERTIFICATIONS

Languages: Python, Java, TypeScript, JavaScript, SQL, OCaml, HTML/CSS
ML & AI: PyTorch, transformers, spaCy, NLP, federated learning, differential privacy, Claude Code, MCP
Web & Infra: FastAPI, Flask, React, Next.js, PostgreSQL, Redis, AWS (EC2), Docker, Git, GitHub Actions, pytest
Security: Splunk, RBAC, MFA, JWT auth, mTLS, encryption, risk registers
Certification: Claude Certified Architect – Foundations, Anthropic (Jul 2026)

LEADERSHIP & ACTIVITIES

- **IEEE/ACM Student Chapter**, University & External Relations Committee Chair — Mar 2026 – Present
- **CyberHawks**: National Cyber League (Spring 2026), team top 4%, individual top 14% — Sep 2025 – Present